

BARNSTABLE, MA, USA

WMO: 725067

Lat: 41.669N

Lon: 70.280W

Elev: 17

StdP: 101.12

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 94720

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.3	-9.7	-21.5	0.6	-10.6	-18.6	0.7	-7.7	13.1	0.2	11.8	1.9	4.1	300	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	29.2	22.9	27.7	22.1	26.5	21.5	24.4	27.5	23.7	26.2	22.9	25.2	5.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.4	18.2	26.0	22.8	17.5	25.3	22.3	17.0	24.7	73.6	27.4	70.6	26.3	68.0	25.1	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.5	8.5	-16.0	32.3	3.1	2.1	-18.2	33.8	-20.0	35.1	-21.8	36.2	-24.0	37.8		
(5)			-16.8	25.8	3.0	1.0	-19.0	26.5	-20.7	27.0	-22.4	27.6	-24.6	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-0.6	0.0	2.9	7.9	13.2	18.2	22.1	21.5	17.7	12.1	7.0	2.5
	DBStd	8.81	5.17	4.40	4.14	3.44	3.54	3.50	2.78	2.91	3.49	4.05	4.29	4.55
	HDD10.0	1286	328	279	221	82	9	0	0	0	0	24	109	234
	HDD18.3	3188	586	512	477	314	164	45	4	6	53	198	339	490
	CDD10.0	1448	0	0	2	17	108	246	374	357	233	88	19	3
	CDD18.3	309	0	0	0	1	4	41	120	105	35	3	0	0
	CDH23.3	1512	0	0	0	5	28	174	656	543	100	6	0	0
	CDH26.7	272	0	0	0	1	6	35	134	86	9	1	0	0
Wind	WSAvg	4.1	4.4	4.4	4.7	4.4	4.1	3.9	3.8	3.6	3.6	4.0	4.1	4.1
Precipitation	PrecAvg	1113	91	92	90	100	93	81	77	87	86	95	115	107
	PrecMax	1573	194	232	242	198	232	195	254	258	263	210	221	234
	PrecMin	699	30	26	24	26	16	5	16	6	6	36	16	15
	PrecStd	196	48	48	47	48	54	54	55	58	54	46	50	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	12.4	16.4	21.2	27.1	30.2	32.1	30.9	27.8	24.0	18.1	15.4
		MCWB	11.0	9.8	10.9	15.0	18.9	22.5	24.4	24.0	22.1	20.0	15.5	13.4
	2%	DB	10.3	10.0	12.6	17.0	22.9	27.3	29.7	28.7	25.8	21.7	16.8	13.1
		MCWB	9.2	8.3	8.9	11.9	17.1	20.9	23.1	23.1	21.1	18.6	14.7	11.8
	5%	DB	8.7	8.3	10.9	14.9	20.9	25.1	27.9	27.4	24.2	20.0	15.4	11.8
		MCWB	7.3	6.3	8.5	10.8	15.8	19.7	22.2	22.5	20.6	17.6	13.3	10.6
	10%	DB	7.2	7.0	9.0	13.2	18.8	23.6	26.6	26.2	23.0	18.4	13.9	10.0
		MCWB	5.7	5.1	6.9	9.8	14.7	19.0	21.6	21.8	20.2	16.2	12.0	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	11.1	12.5	15.7	20.3	23.6	25.6	25.5	23.9	21.1	16.8	14.1
		MCDB	11.5	11.6	14.1	19.1	24.7	27.9	30.6	29.4	26.3	22.6	17.7	15.0
	2%	WB	9.7	8.7	10.4	13.6	18.3	22.3	24.4	24.6	22.8	19.4	15.4	12.3
		MCDB	10.2	9.6	11.7	15.8	21.6	25.7	28.0	27.3	24.5	20.9	16.5	12.9
	5%	WB	7.8	7.0	8.8	12.2	16.9	21.2	23.7	23.9	21.9	18.1	13.9	10.7
		MCDB	8.5	8.0	10.4	13.7	19.4	23.8	26.5	26.1	23.3	19.4	15.0	11.4
	10%	WB	5.9	5.2	7.3	10.8	15.7	20.2	22.9	23.1	21.0	16.8	12.4	8.7
		MCDB	7.0	6.7	8.6	12.5	18.1	22.6	25.2	25.1	22.6	18.1	13.5	9.8
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.1	7.7	8.2	8.1	7.8	7.7	7.9	8.4	8.6	8.1	7.9
		MCDBR	8.8	9.6	9.5	10.5	10.7	9.6	8.8	8.1	8.2	9.0	8.3	8.9
		MCWBR	8.5	8.4	7.5	6.7	6.0	5.1	4.4	4.3	5.3	6.7	7.5	8.9
	5% WB	MCDBR	8.6	9.4	8.7	8.4	9.3	8.1	7.3	6.8	6.7	8.0	7.4	8.7
		MCWBR	8.9	9.0	7.9	6.5	5.8	4.9	4.1	4.0	5.1	6.9	7.4	9.2
Clear-Sky Solar Irradiance	taub	0.310	0.315	0.336	0.375	0.409	0.461	0.484	0.440	0.374	0.347	0.319	0.307	
	taud	2.516	2.503	2.480	2.394	2.342	2.230	2.175	2.292	2.456	2.532	2.567	2.565	
	Ebn at Noon	846	898	914	896	868	821	796	823	862	849	829	821	
	Edh at Noon	73	85	98	114	124	139	145	125	99	82	69	64	
All-Sky Solar Radiation	RadAvg	1.73	2.63	3.67	4.76	5.38	5.77	5.96	5.37	4.36	2.96	1.97	1.47	
	RadStd	0.14	0.25	0.28	0.51	0.51	0.52	0.53	0.33	0.28	0.31	0.25	0.14	0.15

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+96	+45

Nomenclature: See separate page